

Tinkering & Mentoring - VACC 201



Final Project Presentation & Demonstration



Cleanzyme – Bioenzyme Cleaner

Date

(01/04/2026)

Name of the Presenter: Rim Shaikh, Keshvi Sharma, Disha Sikligar,

~~25sc02075@gsfcuniversity.ac.in~~

25sc02076@gsfcuniversity.ac.in

25sc02079@gsfcuniversity.ac.in

Mobile No.:

~~9523670896~~

~~25sc02089@gsfcuniversity.ac.in~~

Faculty Mentor Name: Dr. Sejal Rathod

Team No.:

SOS_13

Semester: 2nd

Program: B.Sc.

Biotechnology
School: School of Science

Student Team

Name	Enrollment No.	Key Responsibility
Rim Shaikh	25SC02075	Idea identification and presentation
Keshvi Sharma	25SC02076	Deep Research and presentation
Disha Sikligar	25SC02079	Prototype formulation and testing
Keya Sojiya	25SC02080	Presentation

Faculty Mentor

Name	Designation	Department, School
Dr. Sejal Rathod	Assistant Professor	Life Sciences, School of Science



Introduction of Tinkering Project

Title of Tinkering Project: Cleanszyme-Bioenzyme Cleaner

Sector: Green Biotechnology / Sustainable Innovation

Brief:

- Natural cleaning solution made from citrus peels, jaggery and water through fermentation.
- Eco-friendly alternative to chemical cleaners.
- Used for odor removal, surface disinfection and as an alternative for bio detergent.

Reference: Ration Bio – Bioenzyme Cleaning Solutions, EnzymEra – Bioenzyme Cleaner Information

Methodology and Tinkering Process

Project Approach:

1. Collection of fruit peels
2. Fermentation (3 months)
3. Filtration and Enzyme extraction
4. Product testing

Tinkering Journey:

- Tested ingredient ratios
- Improved fermentation conditions
- Optimized cleaning efficiency through mentor feedback

Reference: *What Is an Enzyme Cleaner?* , *Natural Bio Enzyme Cleaner Preparation*

Tools and Technology Used

Tools Used and Their Usefulness

1. Fermentation Containers (Airtight Plastic Bottles/Drums)

- Used to carry out anaerobic fermentation of citrus peels, jaggery, and water.
- Useful for producing bioenzymes through microbial activity.

2. pH Meter / pH Strips

- Used to monitor acidity during fermentation (pH = 3-4).
- Useful to ensure proper enzyme formation and product stability.

3. Filtration Setup (Sieve/Filter Paper)

- Used to separate liquid bioenzyme from solid residues.
- Useful for obtaining purified cleaner solution.

4. Measuring Cylinders/Weighing Balance

- Used for maintaining correct ratios of ingredients (1:3:10).
- Useful for formulation accuracy and reproducibility.

5. Spray Bottles / Storage Bottles

- Used for packaging and testing the final product.
- Useful for easy application and prototype demonstration.

Reference: *Bioenzyme Cleaning Technology – Ration Bio* , *Bio-Gleamy Eco-Friendly Bioenzyme Cleaner*

Tools and Technology Used

Technology Used

Fermentation Technology

- Converts organic waste into enzyme-rich cleaner through microbial action.
- Useful because it produces a sustainable, low-cost and biodegradable cleaning agent.

Biodegradation Concept

- Enzymes break down grease, stains and organic dirt naturally.
- Useful for chemical-free cleaning and pollution reduction.

Green Biotechnology Approach

- Uses biological processes for eco-friendly product development.
- Useful for waste management and sustainable innovation.

Overall Utility: These tools and technologies help in low-cost production, quality testing and development of an effective eco-friendly cleaner.

Reference: : *Bioenzyme Cleaning Technology – Ration Bio* , *Bio-Gleamy Eco-Friendly Bioenzyme Cleaner*

Project Outcome & Development

Developed Outcome (Prototype)

- Developed **Cleanszyme**, a natural bioenzyme-based cleaner made from citrus peels, jaggery and water through fermentation.
- Prototype prepared as a multipurpose eco-friendly cleaning solution.

Development Process

Step 1: Idea Identification

- Identified need for a chemical-free, sustainable cleaner using kitchen waste.

Step 2: Formulation Development

- Prepared mixture in **1:3:10 ratio**
(Fruit peels : Jaggery : Water)

Step 3: Fermentation Process

- Allowed fermentation for about **90 days** for enzyme production.
- Regular monitoring of pH, gas release and odor.

Reference: *Organic Recycled Bio Enzyme Natural Floor Cleaner, Cizar Bio-Enzyme Cleaning Products*

Project Outcome & Development

Step 4: Optimization (Tinkering Process)

- Tested different peel combinations and ingredient ratios.
- Improved cleaning efficiency through repeated trials and mentor feedback.

Step 5: Prototype Testing

- Tested on stain removal, odor control and surface cleaning.

Challenges Faced

- Long fermentation time
- Controlling contamination during fermentation
- Standardizing enzyme concentration
- Improving cleaning efficiency compared with commercial cleaners

Reference: : *Organic Recycled Bio Enzyme Natural Floor Cleaner, Cizar Bio-Enzyme Cleaning Products*

Project Outcome & Development

Use Cases

- Floor and kitchen cleaner
- Drain cleaner
- Odor remover
- Organic pesticide for plants

Outcome Achieved

- Low-cost biodegradable cleaner developed
- Converts waste into useful product
- Sustainable alternative to chemical cleaners

Reference: : *Organic Recycled Bio Enzyme Natural Floor Cleaner, Cizar Bio-Enzyme Cleaning Products*

Prototype Image



Day-12



Day-20



Day-30



Final Product



That thin, smooth, slightly wrinkled white layer is kahm yeast.



What you're seeing at the bottom of the jar is normal fermentation sediment.

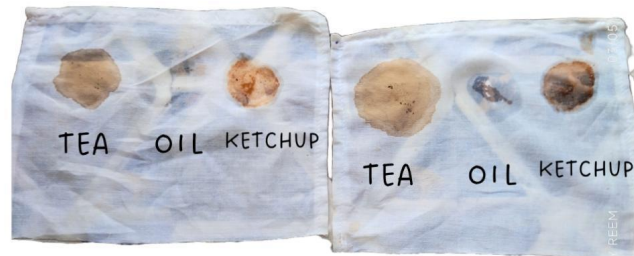


What you're seeing on top is fermentation foam / CO₂ bubbles.

Stain Test

A = Detergent for example wheel, Ariel, Tide etc

B = Bioenzyme Cleaner CleanZyme (prototype)



A B
FRESH STAIN



A B
AFTER 3 HOURS



A B
AFTER WASH



Stain after 3 hours

Comparison Table

Feature	Chemical Detergents	Market Bio-Cleaners	Our Bio-enzyme Cleaner
Composition	Synthetic chemicals	Enzymes + chemicals	Orange peels, jaggery & water
Safety	May irritate skin	Moderately safe	Non-toxic & skin-friendly
Environmental Impact	Causes pollution	Partially eco-friendly	Fully biodegradable & eco-friendly
Odor Control	Artificial fragrance	Moderate	Natural citrus freshness
Cost	Expensive	Moderate	Low-cost preparation
Sustainability	Chemical-based	Semi-natural	Converts kitchen waste into useful product
Residue	May leave chemical residue	Less residue	Residue-free cleaning
Innovation Value	Conventional product	Existing market product	Sustainable "Waste to Wealth" innovation

Learning Outcome

- Learned fermentation technology
- Improved problem-solving skills
- Developed sustainable product design thinking
- Understood teamwork and mentoring importance
- Gained innovation and prototype development experience

Reference: What Is an Enzyme Cleaner?

Conclusion & Future Scope

Conclusion:

- Cleanzyme is low-cost, biodegradable and eco-friendly.
- Reduces chemical pollution and converts waste into useful product.

Future Scope:

- Commercial packaging
- Improved enzyme concentration
- Large-scale production
- Multi-purpose organic cleaning products

Reference: *Punjab Agricultural University Bioenzyme Initiative , Ration Bio Sustainability Approach*

Do you want to apply for PoC/Prototype Development support under Students Startup and Innovation Policy (SSIP 2.0) under GUIITAR Council?

No, currently not applying for PoC/Prototype Development support under SSIP 2.0.

➤➤ **Thank You..**